- *Swarm and cooperative autonomy*: The introduction of swarm coordination algorithms allowed groups of drones to function collaboratively, enhancing mission adaptability and resilience.
- *Stealth and low-observable design enhancements*: Fourth-generation drones feature advanced radar-absorbing materials, blended-wing configurations, and infrared signature reduction technologies, ensuring survivability in high-threat environments.
- *5G and beyond-line-of-sight (BLOS) communication*: The adoption of high-bandwidth, encrypted communication systems enabled seamless coordination among multiple UAVs and command centers over long distances.
- *Multi-domain and modular architecture*: UAVs evolved into modular platforms capable of performing various missions ISR, combat, refueling, logistics, or communication relays, by simply reconfiguring payloads and software modules.
- *Cognitive mission management*: Advanced mission computers now allow drones to adapt in real time to changing conditions, adjusting flight parameters, target priorities, or formations autonomously.

This evolution reflects the global defense and industrial trend toward human– machine teaming (HMT), network-centric operations, and intelligent aerial ecosystems, where UAVs act not merely as assets but as independent tactical collaborators in modern missions.

5.6.2 Key pioneers and notable drone models

The rise of fourth-generation drones is the result of sustained innovation by global aerospace pioneers, each contributing unique design philosophies and technologies. The following models, presented in Table 5.1, exemplify the diverse technological advancements, operational versatility, and